

Luís Carlos F. Latoski

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Summary

Physicist trained in statistical mechanics and complex systems, working since May 2025 as a domain expert on the training and evaluation of frontier AI models. The work covers research-grade tasks in physics, chemistry and mathematics: writing and solving problems, reverse engineering model reasoning to expose failure modes, verifying generated code, and leading the review of tasks produced by other experts. It builds on five years of computational research in C, C++ and Python and three peer-reviewed papers, in *Physical Review E* and *Science Advances*.

AI training experience

Mercor

Research Expert and Team Lead

Remote

May 2025–Present

- Research-grade evaluation of model outputs in physics and complex systems, with technical documentation and specialist contributions to training datasets.
- Reviewer lead for expert-written tasks across physics, chemistry, mathematics and biology.
- Team lead for a video question-and-answer set, leading both task production and review.

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Physics Specialist

Remote

May 2025–Present

- Evaluation and improvement of frontier-model outputs in physics and complex systems, across thousands of tasks.
- Prompt and golden-answer pairs for code generation in C, C++, JavaScript and Python.
- Verifiers for generated code and physics simulations, including MuJoCo.
- Concept decomposition, synthetic data generation and model validation.

Outlier

Physics Expert

Remote

May 2025–Apr 2026

Model output evaluation and synthetic data generation in physics and quantitative reasoning.

Research experience

Universidade Federal do Rio Grande do Sul

PhD in Physics

Porto Alegre

2022–2026

Defense scheduled for October 2026. Persistent voter model: C++ agent-based simulation of opinion coarsening with memory. Metabolic networks: network models of intercellular cross-feeding in Python and C++.

Universidade Federal do Rio Grande do Sul

MSc in Physics

Porto Alegre

2020–2022

Opinion formation with self-induced zealots.

Universidade Federal do Rio Grande do Sul

BSc in Physics

Porto Alegre

2012–2019

Publications

2025: L. C. F. Latoski, A. De Martino, D. De Martino. *Cross-feeding percolation phase transitions of intercellular metabolic networks*. *Science Advances* **11**, eadv8216 (2025).

2024: L. C. F. Latoski, W. G. Dantas, J. J. Arenzon. *Opinion inertia and coarsening in the persistent voter model*. *Phys. Rev. E* **109**, 054115 (2024).

2022: L. C. F. Latoski, W. G. Dantas, J. J. Arenzon. *Curvature-driven growth and interfacial noise in the voter model with self-induced zealots*. *Phys. Rev. E* **106**, 014121 (2022).

Skills

AI training: Concept decomposition, synthetic data generation, prompt and golden-answer design, code verification, model validation, task review, technical documentation

Programming: Python (NumPy, SciPy, NetworkX), C, C++, JavaScript, OpenMP, $\LaTeX$, Git

Domains: Statistical mechanics, network theory, opinion dynamics, percolation, metabolic systems

Languages: Portuguese (native), English (fluent)